

RESEARCH OPPORTUNITIES PROPOSAL • HPCX 2026

Optimizing Operating Room Scheduling through Systems Engineering

King Fahad University Hospital (KFHU) • Dammam, Saudi Arabia

King Fahd University of Petroleum & Minerals (KFUPM) • Dhahran, Saudi Arabia

Type of Study: Quantitative Study • Group ID: HPC-40 • Deadline: April – July 2026

Research Team

Name	Phone	Email	Position	Sign
Dr. Majed Alabdali	0534347474	mmalabdali@iau.edu.sa	PI - Medical Director at KFHU	
Abdulmohsen Alshayeb	0590933943	Abdulmohsen.Alshayeb@gmail.com	Team Leader - College of Medicine	
Hawra Abdullah Alnasser	0564525475	hoho.alnasser@gmail.com	Member – College of Medicine	
Fatima Alhamoud	0569687981	tooma2006@icloud.com	Member – College of Medicine	
Mohammad Alshayeb	0540967731	madoroy72@gmail.com	Member – College of Medicine	
Noor Aljubara	0553833458	nooraljobarh@gmail.com	Member – College of Medicine	

1. Background

1.1 Overview and Significance of the Research Topic

The operating room (OR) is among the most strategically critical and financially intensive units within any hospital system. Surgical services account for approximately 40% of hospital revenue while simultaneously generating a comparable share of operational costs [1,2]. This means that even marginal improvements in scheduling efficiency yield significant clinical and economic benefits. Despite this importance, OR scheduling remains one of the most persistently complex challenges in healthcare operations management, requiring the simultaneous coordination of physical spaces, human resources, specialised equipment, anaesthesia services, and post-anaesthesia care unit (PACU) capacity, all subject to hard time constraints and frequent disruptions [3,4].

These interdependencies generate cascading effects: a single deviation, an overrun surgical case, an absent surgeon, or an unavailable mobile device, propagates delays throughout the entire surgical day, increasing idle time, overtime costs, and patient waiting times [5]. For patients, the downstream consequences include delayed procedures, extended hospital stays, and compromised surgical outcomes. For institutions, the financial and reputational toll is equally severe.

In the Kingdom of Saudi Arabia (KSA), these challenges are amplified by the demands of a rapidly expanding healthcare system. The Saudi Vision 2030 Health Sector Transformation Programme explicitly mandates improvements in hospital efficiency, capacity utilisation, and quality of care. King Fahad University Hospital (KFUH) in Dammam, a major public academic hospital affiliated with Imam Abdulrahman Bin Faisal University, operates within these strategic imperatives, making it both an ideal and highly representative site for a rigorous scheduling optimization study. A reproducible, data-driven framework developed at KFUH carries direct implications for system-wide reform across KSA public health institutions.

1.2 Research Question

Can a deterministic Mixed-Integer Linear Programming (MILP) model, parameterised using real historical surgical data from King Fahad University Hospital, generate daily surgical schedules that demonstrably reduce OR idle time and overtime compared to conventional heuristic-based scheduling, while explicitly accounting for shared mobile equipment constraints?

This overarching question is operationalised through three subsidiary inquiries:

- What are the dominant sources of OR inefficiency at KFUH, as revealed by historical scheduling data?
- To what extent does explicitly modelling shared equipment constraints alter schedule quality relative to models that omit this dimension?
- How do MILP-generated schedules perform across a range of synthetic operational scenarios designed to stress-test the model under varying demand and resource conditions?

1.3 Review of Relevant Literature and Identification of Gaps

OR scheduling research has grown substantially over the past two decades. Cardoen, Demeulemeester, and Beliën (2010) [1] established the canonical problem taxonomy and confirmed Mixed-Integer Linear Programming (MILP) as the dominant exact methodology, with minimisation of OR overtime and maximisation of utilisation as the two most common objective functions, objectives that are also central to the present study. Pulido et al. (2014) [9] applied MILP to daily scheduling in a Spanish public teaching hospital, incorporating surgeon skill heterogeneity and simulation-based visualisation, but did not model shared equipment constraints, a gap directly addressed here. Spratt and Kozan (2021) [10] proposed an integrated rolling-horizon MILP approach for medium-term planning, demonstrating strong performance for managing non-elective arrivals, though their tactical planning focus limits direct transferability to daily operational scheduling.

On the stochastic side, Celik et al. (2021/2023) [11] introduced a two-stage Stochastic Mixed-Integer Programming model for hospitals with parallel anaesthesia induction rooms, demonstrating that explicitly modelling duration uncertainty reduces idle time and smooths patient flow. Addis, Carello, and Tanfani (2024) [12] developed a cardinality-constrained robust model incorporating patient urgency weights, highlighting the trade-off between robustness and efficiency. Metaheuristic approaches have also gained traction: Lin and Li (2021) [13] showed that a specialised Artificial Bee Colony algorithm can schedule 150 cases in under seven minutes, while Vieira et al. (2025) [14] combined genetic algorithms with Q-Learning for superior benchmark performance.

Human factors add a further dimension: Liao, Su, and Liu [15] demonstrated that equitable surgeon workload distribution reduces burnout risk without compromising efficiency, and Al-Araidah et al. (2025) [16] integrated queuing-based capacity planning with goal programming to produce resilient, equitable schedules. From a Systems Engineering perspective, Nakka and Kumar (2025) [8] demonstrated that embedding ISE methodologies into surgical workflows yields measurable improvements across multiple performance dimensions. Contextually, Alzahrani et al. (2025) [20] documented a 31% increase in surgical volume and 27% reduction in waiting times at a Saudi hospital following extended OR hours, confirming the capacity pressures motivating this work.

Synthesising this literature, three substantive gaps directly motivate the present study:

- **Shared mobile equipment constraints** are absent from most published MILP models. Equipment is typically treated as room-fixed or unlimited, an assumption incompatible with hospitals where a limited inventory of mobile devices (C-arms, laparoscopic towers, microscopes) must serve multiple concurrent ORs simultaneously [6,7].
- **There is a near-total absence** of OR scheduling studies within the Saudi or GCC public health context. The operational demands, resource configurations, staffing structures, and policy environments of Saudi hospitals differ substantially from the European and North American centres that dominate the literature.
- **The gap between model development** and clinical deployment remains wide [17]. Most studies use synthetic data and produce outputs misaligned with administrative workflows. This study bridges that gap by grounding every model parameter in real KFHu data and producing outputs compatible with existing scheduling practice.

1.4 Significance and Contribution to the Field

This research makes four distinct and original contributions to the literature on healthcare operations management and OR scheduling.

First, it introduces explicit mobile equipment-sharing constraints into the MILP formulation, the primary structural gap in existing models. Second, it is grounded in 135,971 real surgical records from KFUH, enabling parametrisation that reflects the actual case-mix, surgeon availability patterns, room configurations, and equipment inventory of a major Saudi public hospital. Third, by situating the work within Saudi Vision 2030, it establishes a concrete link between mathematical optimization methodology and national health policy, contributing to the evidence base for data-driven healthcare reform across the GCC region. Fourth, adopting a Systems Engineering (SE) perspective treats the OR as a subsystem embedded within the broader hospital architecture capturing interdependencies between the OR, PACU, and interdepartmental equipment pools that narrowly scoped models systematically overlook [8].

2. Research Objectives

2.1 Primary Aim

To develop, validate, and apply a deterministic Mixed-Integer Linear Programming (MILP) model that generates mathematically optimal daily elective surgical schedules at King Fahad University Hospital, minimising OR idle time and overtime while explicitly respecting shared mobile equipment constraints, and to quantify its performance improvements over conventional heuristic-based scheduling, thereby producing a reproducible, data-driven framework aligned with Saudi Vision 2030 health system transformation objectives.

2.2 Specific Objectives

- Conduct a rigorous exploratory data analysis of 135,971 KFHU surgical records to identify dominant sources of OR inefficiency, characterise case-duration distributions by specialty, map room–specialty compatibility patterns, and derive empirical parameters for the MILP model.
- Formulate and implement a MILP model in Python (Pyomo) incorporating: room assignment and compatibility constraints; intra-room sequencing constraints with mandatory cleanup time; novel mobile equipment conflict constraints preventing concurrent use of shared devices across OR suites; and idle time and overtime quantification constraints.
- Validate the MILP model across eight synthetic operational scenarios, spanning baseline, high-demand, equipment-conflict, high-penalty, and complex-compatibility conditions, and quantify performance improvements in idle time, overtime, utilisation, and total cost relative to simulated heuristic baselines.
- Assess the incremental scheduling value of explicitly modelling shared mobile equipment constraints by comparing model outputs with and without this constraint layer.

- Gather qualitative insights from OR stakeholders to validate operational model assumptions, identify implementation barriers, and inform a roadmap for decision-support integration at KFUH.

2.3 Secondary Objectives

- Establish a standardised set of quantifiable Key Performance Indicators (KPIs) for ongoing OR scheduling performance monitoring at KFUH, enabling longitudinal benchmarking aligned with Vision 2030 metrics.
- Develop a scalable scheduling optimization framework that can be adapted and deployed across other KSA public hospital systems, contributing to the national evidence base for data-driven OR management.
- Produce a data-to-model mapping protocol that serves as a replicable methodology for future OR scheduling studies in resource-constrained public healthcare environments.
- Explore the feasibility of integrating the MILP model into existing hospital information systems as a decision-support module, identifying technical and organisational prerequisites for implementation.

3. Methodology

3.1 Study Setting

This study is conducted at King Fahad University Hospital (KFUH), a major public academic hospital affiliated with Imam Abdulrahman Bin Faisal University in Dammam, Saudi Arabia. KFUH operates multiple operating rooms across a wide range of surgical specialties, including General Surgery, Orthopaedics, Urology, Neurosurgery, Ophthalmology, ENT, Gastroenterology, and Obs/Gyn, under conditions of high and growing patient demand. The hospital relies on coordinated use of operating rooms, multidisciplinary clinical teams, and a shared pool of specialised equipment including mobile C-arms, laparoscopic towers, surgical microscopes, and robotic platforms. The study focuses exclusively on elective surgeries, isolating the combinatorial scheduling problem from the stochastic interruptions of emergency cases, a standard and well-justified approach in foundational OR scheduling research.

3.2 Study Design

This research adopts a mixed-methods design integrating quantitative optimization modelling with qualitative stakeholder inquiry. The quantitative component is the methodological core: it encompasses retrospective exploratory data analysis, MILP model development and validation, and comparative scenario-based computational experiments. The qualitative component, comprising structured field observation, semi-structured interviews, and stakeholder surveys, serves to validate operational model assumptions, surface implementation barriers, and ensure the framework is both clinically realistic and administratively actionable.

This design is explicitly consistent with Systems Engineering (SE) principles, treating the OR as a subsystem within the broader hospital architecture. The study timeline is three months (April–July 2026), structured into three sequential phases: (1) data extraction and exploratory analysis; (2) model development, parametrisation, and computational validation; and (3) qualitative data collection, analysis, and final synthesis.

3.3 Study Subjects, Inclusion and Exclusion Criteria

Quantitative Component (Historical Data)

- Inclusion: All elective surgical cases at KFUH between 24 August 2023 and 28 July 2025 with complete Theatre_In_Time and Theatre_Out_Time records, a valid Episode_No identifier, and an assigned bookable OR resource.
- Exclusion: Emergency and trauma cases (Elective_Or_Emergency = 'Emergency'); records with missing, null, or implausible timestamps (negative or >600-minute durations); non-surgical procedural bookings performed outside dedicated OR suites; and records lacking a valid OR room assignment.

Qualitative Component (Human Participants)

- Inclusion: OR stakeholders with direct involvement in surgical scheduling at KFUH, surgeons across all scheduled specialties, anaesthesiologists, scrub and circulating nurses, OR coordinators, administrative scheduling staff, and biomedical technicians responsible for mobile equipment.
- Exclusion: Staff with fewer than three months of continuous experience at KFUH; personnel without direct scheduling responsibilities; non-clinical support staff.

3.4 Sample Size

Quantitative: The dataset comprises 135,971 historical surgical records from KFUH, representing one of the largest real-world OR scheduling datasets in published KSA research. Following application of inclusion and exclusion criteria, the refined dataset is expected to retain approximately 60–70% of total records, providing a large and statistically robust sample for model parametrisation, exploratory analysis, and baseline estimation.

Qualitative: A purposive sample of approximately 15–25 OR stakeholders will be recruited across all identified stakeholder groups, with sample size guided by thematic saturation, consistent with established practice for semi-structured interview studies in healthcare operations settings.

3.5 Sampling Technique

Quantitative (Purposive Filtering): A criterion-based data selection strategy isolates elective surgical cases and aggregates records to the unique surgical case level using Episode_No and Operation as a composite key, eliminating duplication arising from multi-personnel entries per procedure.

Qualitative (Purposive with Snowball Extension): OR stakeholders are identified purposively to ensure representation across all professional roles involved in surgical scheduling. Snowball sampling will be employed where access to specific subgroups (e.g., senior anaesthesiologists, biomedical staff) is restricted.

3.6 Data Collection Methods

Document and System Review

Historical OR scheduling records, staff rosters, equipment inventories, sterilisation logs, and standard operating procedures (SOPs) will be extracted from KFUH's hospital information system. These records provide the empirical foundation for MILP model parametrisation and baseline performance estimation, following a pre-specified data extraction protocol developed in consultation with the KFUH Medical Director.

Field Observation

Direct observation of OR workflows will be conducted across a minimum of two non-consecutive weekdays per specialty block, capturing real-time room turnover processes, equipment usage and allocation patterns, scheduling deviations relative to planned programmes, and coordination behaviours among clinical teams. A structured observation instrument will ensure systematic and reproducible data capture.

Semi-Structured Interviews

Individual semi-structured interviews of 30–45 minutes will be conducted with purposively selected OR stakeholders. The interview guide will address five domains: (i) current scheduling practices and perceived strengths/weaknesses; (ii) common sources of OR delay and inefficiency; (iii) mobile equipment availability and conflict management; (iv) attitudes toward algorithmic decision support; and (v) barriers to and enablers of scheduling optimisation. All interviews will be audio-recorded and transcribed verbatim.

Structured Surveys

Standardised questionnaires using validated Likert-scale instruments will be distributed to all OR staff to quantify perceptions of workflow effectiveness, scheduling accuracy, technology readiness, and workload burden. Survey data will complement and triangulate interview and observational findings.

3.7 Data Analysis Plan

Quantitative Analysis, Optimization Model Performance

The core quantitative analysis evaluates the MILP model's performance relative to a simulated baseline representing current manual (heuristic) scheduling at KFUH, across eight operational scenarios of varying demand and constraint complexity. The primary KPIs are defined in Table 1 below.

Table 1. Key Performance Indicators for MILP Model Evaluation

KPI	Definition	Significance
Total OR Idle Time	Non-productive room time within the scheduling window	Minimise, direct proxy for wasted capital expenditure
Total Overtime (min)	Cumulative time beyond the standard 07:00–15:00 day	Minimise, controls labour cost and staff well-being
Total Minimised Cost Z	Weighted sum: $\alpha \times \text{Idle Time} + \beta \times \text{Overtime}$	Primary objective function, policy-tunable via $\alpha:\beta$ ratio
OR Utilisation Rate (%)	Proportion of available time occupied by surgical activity	Maximise, balanced against burnout risk
Schedule Feasibility Rate	% of scenarios producing a valid, constraint-satisfying schedule	Robustness indicator across demand conditions
Equipment Conflict Rate	Frequency of unresolvable shared-device scheduling clashes	Target: 0% in all MILP outputs

Statistical significance of KPI differences between MILP-optimised and simulated baseline schedules will be assessed using paired t-tests (continuous, normally distributed KPIs) and Wilcoxon signed-rank tests where normality cannot be assumed. Chi-square tests of independence will evaluate categorical outcomes such as equipment conflict frequency and schedule feasibility rates. A significance threshold of $p < 0.05$ will be applied throughout, with Bonferroni correction for multiple comparisons where appropriate. Sensitivity analyses will examine how model performance responds to variation in the α (idle time penalty) and β (overtime penalty) cost coefficients, informing administrators on the practical consequences of different cost prioritisation policies.

Qualitative Analysis

Interview transcripts and observational field notes will undergo framework-based thematic analysis, conducted by two independent coders to ensure analytical rigour. Inter-rater reliability will be assessed using Cohen's Kappa, with discrepancies resolved through discussion. Survey responses will be analysed descriptively and inferentially, using Chi-square and Mann-Whitney U tests to identify significant differences in perceptions across professional groups. Qualitative and quantitative findings will be integrated in a final synthesis phase to produce actionable implementation recommendations.

3.8 Scheduling Optimization Model

The methodological centrepiece of this study is a deterministic Mixed-Integer Linear Programming (MILP) model implemented in Pyomo, a Python-based open-source algebraic modelling language. A deterministic MILP formulation was chosen over heuristic or stochastic alternatives for three reasons: (i) it provides mathematically proven optimal solutions for a given input set, ensuring administrative accountability and defensibility; (ii) it produces transparent, interpretable outputs, specific start times, room assignments, and cost breakdowns, directly communicable to scheduling staff; and (iii) it establishes a rigorous, reproducible benchmark against which stochastic extensions can be evaluated in future work.

Objective Function

The model minimises a weighted total cost combining the two key drivers of OR inefficiency:

$$\text{Minimise } Z = \alpha \times \Sigma(\text{Idle Time per Room}) + \beta \times \Sigma(\text{Overtime per Room})$$

Where α is the cost coefficient per minute of idle time (unused, paid-for OR capacity), and β is the cost coefficient per minute of overtime (labour cost and staff well-being impact). By adjusting the $\alpha:\beta$ ratio, hospital administrators can tune the model to their specific operational priorities, placing greater weight on overtime reduction during periods of high nursing cost pressure, or on utilisation maximisation during periods of surgical backlog.

Constraint Structure

The model operates on three classes of decision variables: binary room-assignment variables; binary sequencing variables establishing procedure order within each room; and continuous time variables defining precise start and finish times for each case. These are governed by five constraint families:

- **Assignment Constraints:** Each surgery is assigned to exactly one operating room and only to rooms for which it is clinically compatible, based on fixed equipment requirements and specialty–room mappings derived from KFHU data. This ensures every generated schedule is immediately feasible.
- **Timing Constraints:** Each surgery's finish time equals its start time plus its empirically derived procedure duration. This links continuous time variables to the specialty-specific processing time parameters from the dataset.
- **Intra-Room Sequencing Constraints:** For any two surgeries in the same OR, one must be fully completed, including mandatory post-procedure cleanup, before the next can begin. The logical disjunction is linearised using the standard Big-M method, enabling exact solution with open-source MILP solvers.
- **Mobile Equipment Conflict Constraints (*Novel Contribution*):** If two distinct surgeries both require the same type of mobile equipment, for example, a portable C-arm or surgical laser, they cannot be scheduled concurrently, even across different ORs. This constraint set, absent from most published models, ensures schedule feasibility with respect to the hospital's actual mobile device inventory. In preliminary experiments, this constraint automatically resolved conflicts that manual schedulers manage through error-prone ad hoc communication.
- **Idle Time and Overtime Calculation Constraints:** The room closure time is tracked explicitly for each OR. Overtime accumulates whenever closure time exceeds the standard day boundary ($T_{\text{max}} = 15:00$). Idle time is the difference between total room availability and total surgical occupancy, accounting for sequencing gaps and cleanup intervals.

Why MILP Over Manual Scheduling

Manual scheduling at KFUH is a heuristic, experience-based process. While schedulers possess valuable clinical intuition, they cannot systematically evaluate the combinatorial solution space, which grows factorially with case count, cannot simultaneously enforce multi-dimensional constraints such as mobile equipment conflicts across concurrent ORs, and are subject to cognitive biases such as

anchoring on familiar case sequences. The MILP model exhaustively explores the feasible solution space to deliver a mathematically optimal schedule for the defined objective, elevating the scheduler's role from reactive case sequencer to strategic resource planner, with the capacity to re-optimize in minutes in response to intraday disruptions such as emergency add-ons, late cancellations, or equipment unavailability.

3.9 Exploratory Data Analysis and Model Parametrisation

The historical dataset provided by KFHU comprises 135,971 surgical records spanning 24 August 2023 to 28 July 2025. A rigorous exploratory data analysis was conducted to translate this operational data into the structured inputs required by the MILP model. The following key findings, illustrated by the figures below, directly inform the model's parameters and constraint structure.

Dataset Structure and Preprocessing

Each unique surgical event is represented by multiple dataset rows corresponding to different personnel involved. A primary preprocessing step aggregated records to the unique surgical case level using Episode_No and Operation as a composite key, eliminating duplication. Elective cases were then isolated and records with implausible timing values excluded. Table 2 summarises the complete mapping from raw KFHU data fields to MILP model parameters.

Table 2. KFHU Data Field Mapping to MILP Model Parameters

MILP Parameter	Description	KFHU Data Field(s)	Preprocessing Step
Set of Surgeries (S)	Unique elective cases to schedule	Episode_No + Elective_Or_Emergency	Filter 'Elective'; deduplicate by Episode_No
Set of Rooms (R)	Available operating theatres	book_Resource	Extract unique room identifiers
Processing Time (Ps)	OR occupancy time per case	Theatre_In_Time, Theatre_Out_Time	Compute difference: Theatre_Out – Theatre_In
Room Compatibility	Room–surgery assignment feasibility	book_Resource, Operation	Map rooms; flag fixed-equipment requirements
Mobile Equipment	Shared device need per case	Surgeon_Specialty, OPSTFRL_Desc	Identify device type by procedure category
Cleanup Time (C)	Mandatory inter-case turnover	Not recorded, derived empirically	Set 15–25 min from turnover time distribution
Standard Day (T_max)	Overtime onset boundary	Theatre_In_Time peaks	Parameterised as 08:00–15:00 (Sunday–Thursday)

Resource Utilisation and Bottleneck Identification

Analysis of room utilisation across all bookable resources revealed a markedly uneven distribution with a pronounced long-tail pattern. A small number of ORs, notably OR 5, OR 10, and OR 7, handle a disproportionately high surgical volume, identifying them as potential system bottlenecks requiring priority attention in scheduling optimisation. Figure 1 displays the full usage frequency ranking.

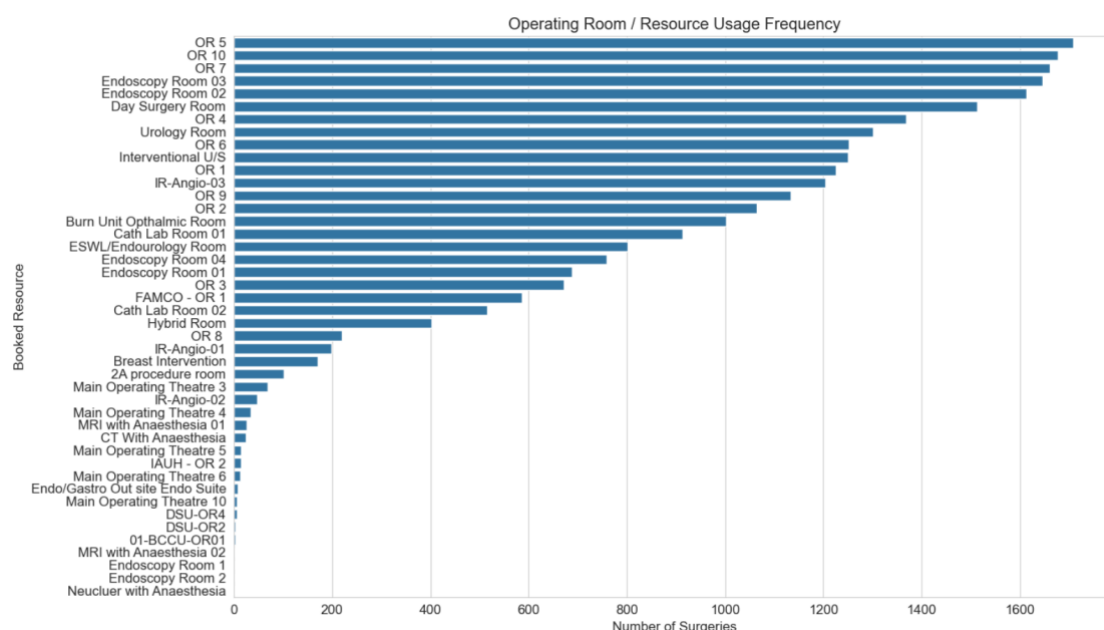


Figure 1. OR / Resource Usage Frequency across all bookable rooms, ranked by surgical volume (KFHU 2023–2025).

A comparable caseload concentration is observed by surgeon specialty (Figure 2). Consultant Gastroenterology and Consultant Urology manage the highest case volumes by a substantial margin, while specialties such as Consultant Neurosurgery handle far fewer but substantially longer and more variable procedures. This heterogeneity directly motivates the use of specialty-specific processing time parameters in the model.

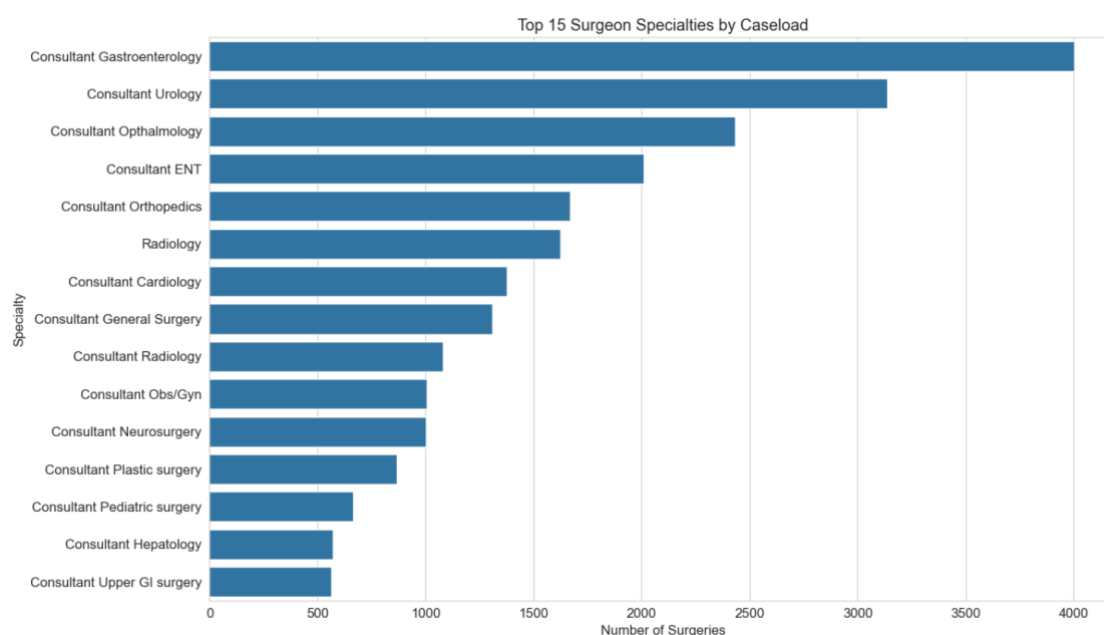


Figure 2. Top 15 Surgeon Specialties by total caseload (KFHU 2023–2025).

Operational Window Parametrisation

Analysis of surgical volume by day of week (Figure 3) confirms that KFUH operates a standard Sunday-to-Thursday surgical week, with negligible activity on Fridays and Saturdays. This establishes the planning horizon for the model. Analysis of surgery start times by hour (Figure 4) confirms that peak operational onset is 08:00, with activity tapering after 15:00. These two findings together parametrise the standard day boundary $T_{\max} = 15:00$, the threshold beyond which overtime penalties begin to accrue in the objective function.

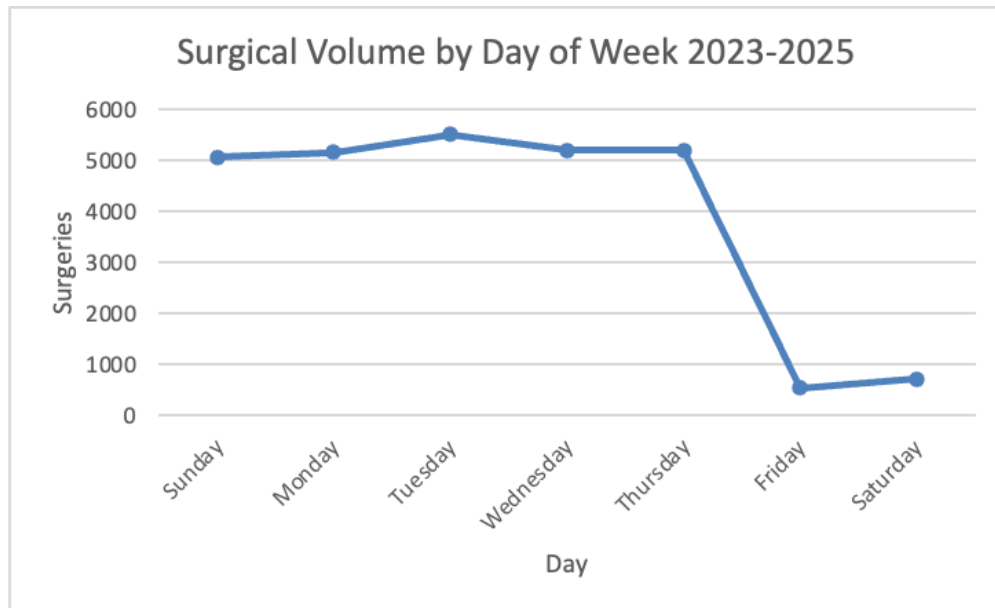


Figure 3. Surgical Volume by Day of Week confirming the Sunday–Thursday operational week and negligible weekend activity.

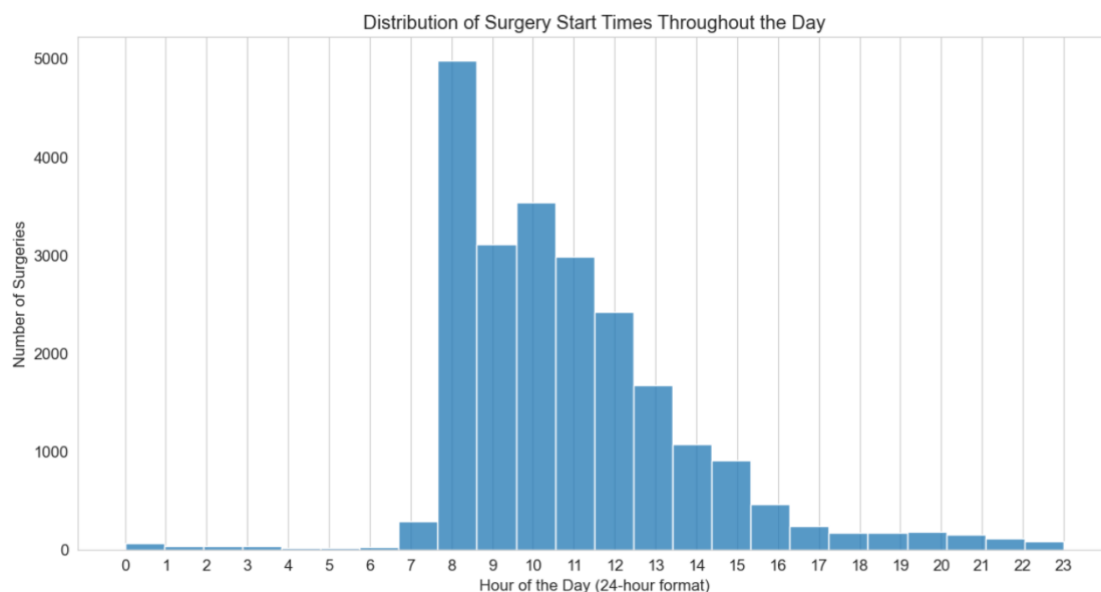


Figure 4. Distribution of Surgery Start Times (24-hour format) establishing 08:00 as the primary operational onset and 15:00 as the standard day boundary.

Surgical Duration Distribution and Specialty-Level Variability

Figure 5 presents boxplots of surgical duration disaggregated by the top 10 surgeon specialties. The variation is stark: Consultant Neurosurgery and Consultant Orthopaedics procedures are not only substantially longer on average but also far more variable than Consultant Gastroenterology or Consultant Ophthalmology cases. This finding directly validates the use of specialty-specific processing time parameters (Ps) in the model, a critical design choice that significantly improves schedule quality over models using a uniform average procedure time. The pronounced positive skew in the overall duration distribution confirms that a simple mean would be a highly inaccurate scheduling input.

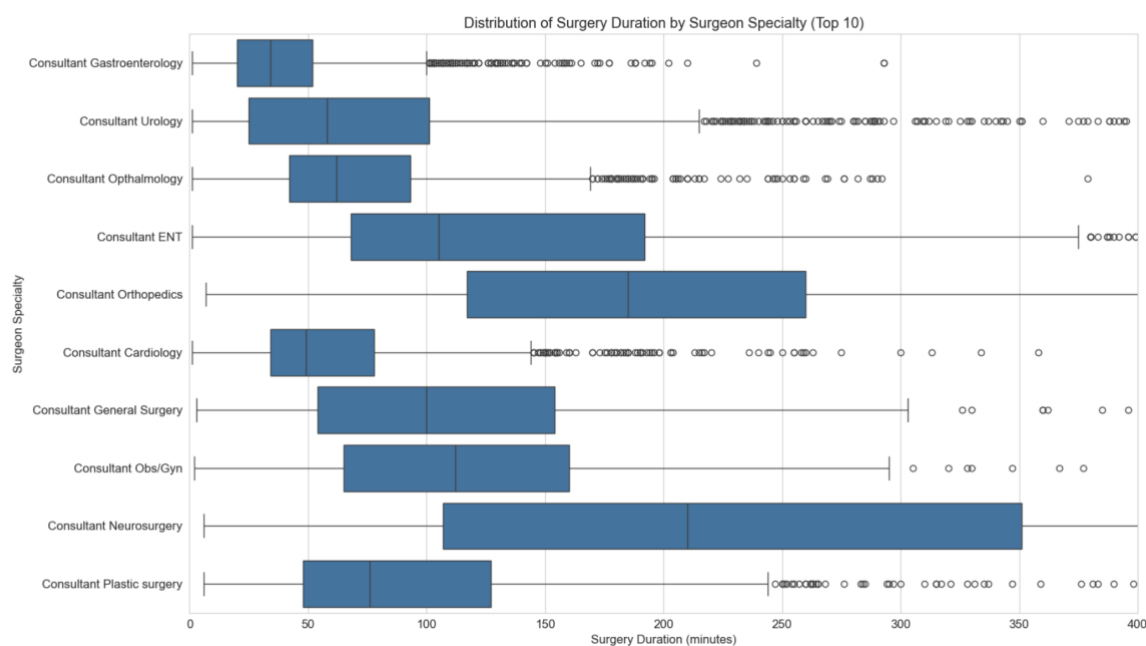


Figure 5. Boxplots of Surgery Duration by Top 10 Surgeon Specialties, illustrating the stark inter-specialty variation in both mean duration and variability.

OR Turnover Time Distribution

Figure 6 displays the distribution of inter-procedural OR turnover times, filtered for plausible gaps (5–120 minutes). The distribution is right-skewed with a mode of approximately 10–15 minutes and a mean of approximately 20–25 minutes. This empirical distribution provides the basis for setting the mandatory cleanup time parameter (C) in the model's sequencing constraints, replacing the arbitrary assumptions common in the theoretical literature with data directly measured from KFUH operations.

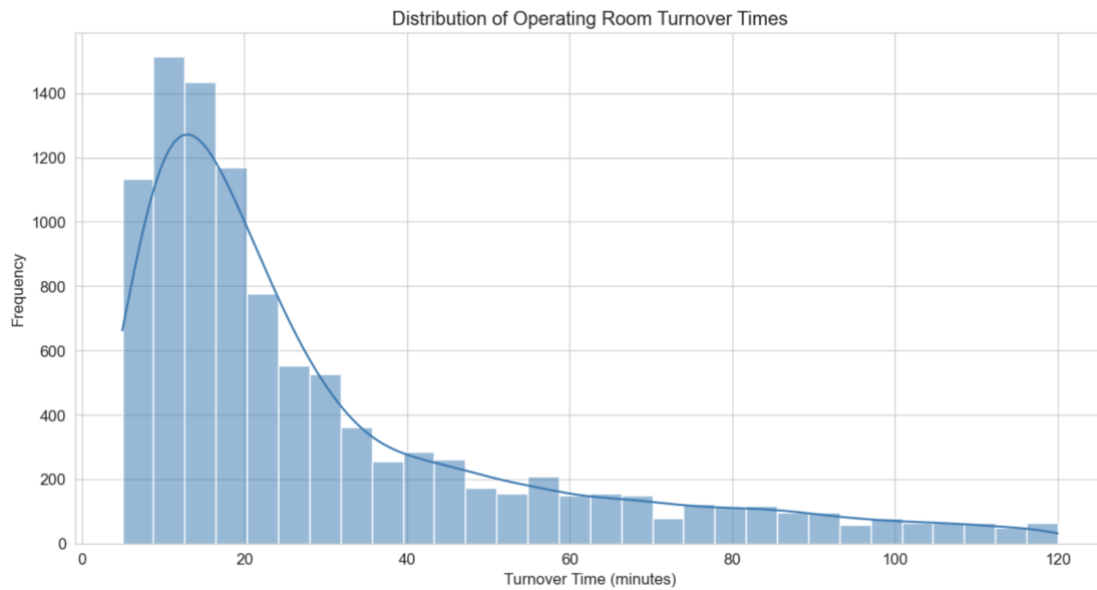


Figure 6. Distribution of OR Turnover Times (5–120 min), providing the empirical basis for the model's mandatory cleanup time parameter.

Room Specialty Compatibility Mapping

Figure 7 presents a heatmap cross-tabulating surgeon specialties against operating rooms. The sparse, non-random pattern is striking: Consultant Ophthalmology almost exclusively uses the Burn Unit Ophthalmic Room; Consultant Orthopaedics shows a dominant affinity for OR 1; Consultant Urology concentrates in the Urology Room and Day Surgery Room. These clear, data-driven pairings confirm that many room–specialty assignments are effectively hard constraints dictated by fixed equipment and clinical practice. They directly populate the room compatibility parameters in the model's assignment constraints, ensuring all generated schedules are clinically feasible.



Figure 7. Heatmap of Caseload by Surgeon Specialty and Operating Room, confirming strong non-random specialty–room associations that populate the model's compatibility constraints.

Preliminary Optimization Results: Eight Scenario Analysis

The MILP model was validated against eight synthetic operational scenarios specifically designed to stress-test its robustness across the full range of demand and constraint conditions expected at KFUH. Each scenario was constructed to isolate a specific operational challenge, enabling systematic assessment of model behaviour. The model successfully generated proven-optimal schedules in all eight completed scenarios. Table 4 summarises the quantitative outcomes, and Figures 8 and 9 provide visual decomposition of performance.

Table 4. MILP Optimization Results Across Eight Operational Scenarios

#	Scenario Name	Cost (Z)	OT (min)	Idle (min)	Description & Key Finding
1	Baseline Extension	600	0	60	Standard 3-OR elective day. Serves as performance benchmark for all subsequent scenarios.
2	High Demand & Limited Capacity	4,100+	110+	80+	Full case backlog in 3 ORs; demand exceeds capacity. Model demonstrates cost of system overload.
3	Fixed Equipment Bottleneck	600	0	60	One OR has a fixed C-arm usable only for specific specialties. Model respects hard compatibility constraints.
4	High Overtime Penalty ($\beta \gg \alpha$)	150	0	30	Cost weight heavily penalises overtime. Model aggressively compacts cases to stay within standard hours.
5	High Idle Time Penalty ($\alpha \gg \beta$)	5,000	0	50	Cost weight heavily penalises idle time. Model densely packs rooms; accepts minor gaps over any overtime.
6	Long & Complex Procedures	300	0	30	Mix of very long neurosurgery/orthopaedics cases. Model sequences correctly to avoid cascading overtime.
7	Mobile Equipment Conflict	2,325	75	45	Three ORs need same mobile laser simultaneously. Model staggers them to eliminate conflicts automatically.
8	Complex Multi-Room Compatibility	600	0	60	Multiple specialty-room restrictions active simultaneously. Model navigates all constraints feasibly.

Scenario Descriptions and Key Findings

Scenario 1, Baseline Extension: A standard three-OR elective day with moderate case volume and typical duration variability. This scenario serves as the primary performance benchmark for all subsequent comparisons. The model generated a fully feasible schedule with zero overtime and 60 minutes total idle time across all rooms, at a total cost Z of 600. This baseline demonstrates that the model can handle routine scheduling conditions with high efficiency.

Scenario 2, High Demand and Limited Capacity: A full surgical backlog is scheduled across three ORs, with demand deliberately exceeding available capacity. This scenario tests the model's behaviour under system overload conditions. Total cost exceeded 4,100, driven almost entirely by overtime of over 110 minutes, reflecting the unavoidable spillover when case volume exceeds the standard day. Critically, the model still produced the optimal schedule given the constraints, minimising the damage rather than allowing uncontrolled overtime.

Scenario 3, Fixed Equipment Bottleneck: One OR contains a fixed, non-mobile C-arm usable only by specific surgical specialties. This scenario validates the model's room compatibility constraints. Total cost was 600 with zero overtime, identical to the baseline, demonstrating that the model correctly navigates fixed equipment restrictions without degradation in schedule quality for the assignable caseload.

Scenario 4, High Overtime Penalty ($\beta \gg \alpha$): The cost coefficient for overtime is set substantially higher than for idle time, reflecting an operational priority to protect staff from excessive working hours. The model responds by aggressively compacting cases to ensure all procedures finish within standard hours, achieving zero overtime and the lowest total cost (150) across all scenarios, at the cost of slightly reduced idle time reduction. This demonstrates the model's precision as a policy instrument.

Scenario 5, High Idle Time Penalty ($\alpha \gg \beta$): The cost coefficient for idle time is set substantially higher, reflecting an institutional priority to maximise OR utilisation. The model densely packs rooms to eliminate idle gaps, producing zero overtime but incurring the highest total cost (5,000), entirely from the severe penalty on 50 minutes of residual idle time. This scenario powerfully illustrates that the cost structure of the objective function directly shapes the model's scheduling strategy, enabling administrators to express different management priorities quantitatively.

Scenario 6, Long and Complex Procedures: The caseload includes a high proportion of lengthy, variable-duration procedures typical of Neurosurgery and Orthopaedics. The model correctly sequences these cases to avoid cascading overtime, achieving zero overtime and only 30 minutes total idle time at a cost of 300, demonstrating that specialty-specific duration parametrisation is essential for accurate scheduling of complex surgical days.

Scenario 7, Mobile Equipment Conflict: Three concurrent surgeries in three different ORs all require the same shared mobile laser. This is the critical validation scenario for the novel equipment conflict constraints. The model automatically staggers the three procedures across the day so that no two require the laser simultaneously, a coordination task that manual schedulers manage through

error-prone verbal communication and that is entirely invisible to models lacking this constraint layer. Total cost was 2,325 with 75 minutes overtime, reflecting the scheduling cost of the imposed temporal separation. Without the conflict constraint, the model would have produced an infeasible schedule.

Scenario 8, Complex Multi-Room Compatibility: Multiple specialty-room restrictions are active simultaneously across all three ORs, creating a highly constrained assignment problem. The model navigates all compatibility restrictions and produces a fully feasible schedule with zero overtime and 60 minutes idle time at cost 600, confirming the correct implementation of the compatibility constraint structure and the model's robustness under complex real-world assignment restrictions.

Visual Performance Analysis

Figure 8 presents the three-panel performance overview across all eight scenarios, displaying Total Minimised Cost Z, Total Overtime, and Total Idle Time. Figure 9 decomposes Total Cost Z into its idle-time and overtime components, visually confirming that the model directly responds to the cost structure of each scenario, a property essential for its use as a management decision-support tool.

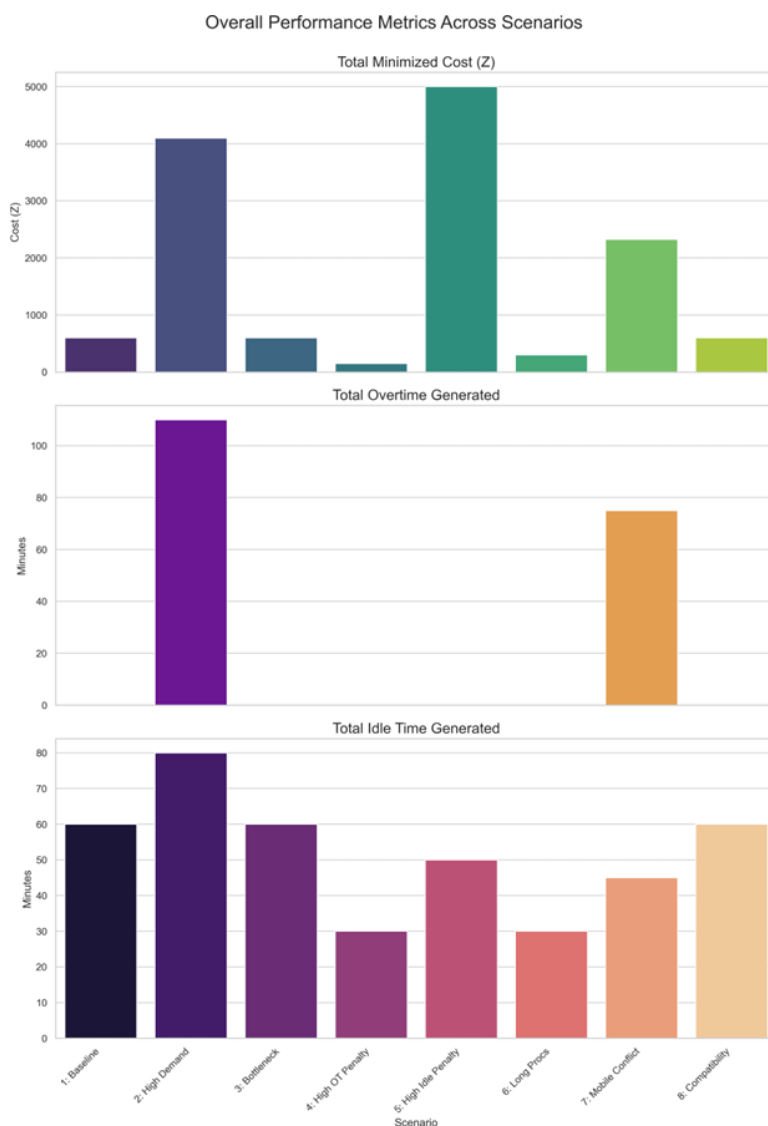


Figure 8. Overall Performance Metrics across all eight test scenarios: Total Minimised Cost Z (top), Total Overtime in minutes (middle), Total Idle Time in minutes (bottom).

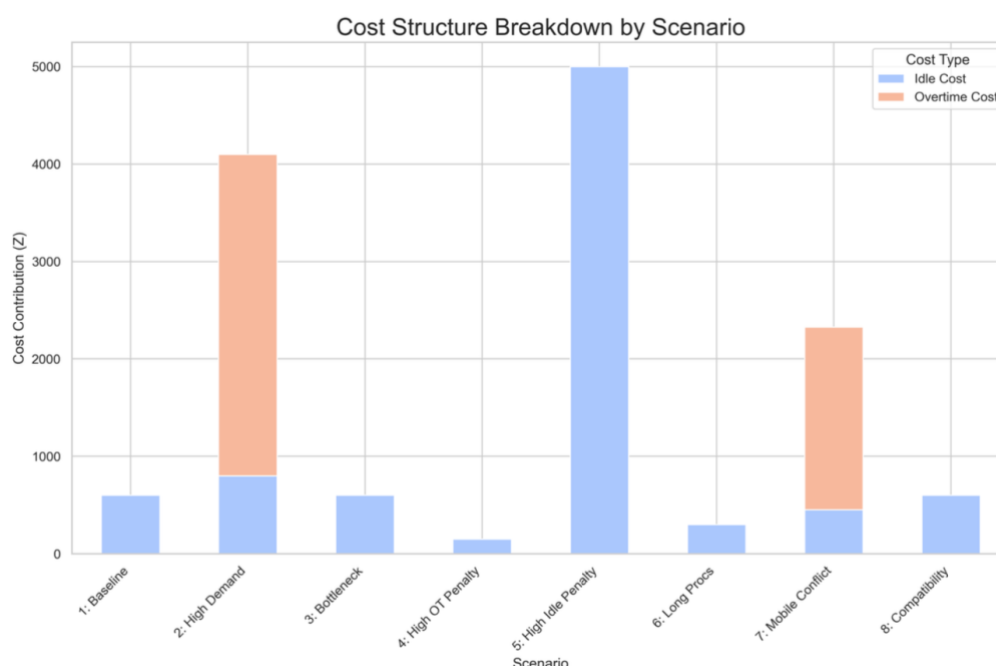


Figure 9. Cost Structure Breakdown by Scenario, decomposing Total Cost Z into its idle-time cost component (blue) and overtime cost component (orange).

4. Ethical Considerations

This study has been approved by the Institutional Review Board (IRB) of King Fahad Hospital University. All research activities involving human participants will be conducted in full accordance with the Declaration of Helsinki and applicable KSA national research ethics guidelines.

Informed consent will be obtained from all qualitative participants prior to data collection, following a structured consent process that clearly explains: study objectives and design; the voluntary nature of participation and the right to withdraw at any time without consequence; how data will be used, stored, and reported; and the confidentiality protections in place. Participation will have no bearing on employment status or professional standing.

All quantitative data extracted from KFUH's information system will be fully anonymised at source, with all direct and indirect patient and staff identifiers removed prior to transfer to the research team. Access to the anonymised dataset will be restricted to the principal investigator, the supervising Medical Director, the designated biostatistician, and authorised data managers. Data will be stored on password-protected, institutionally secured systems for the duration of the study and archived or securely deleted in accordance with IRB-stipulated data retention policies. No conflict of interest exists among any member of the research team.

IRB Approval: Study approved by the Institutional Review Board of King Fahad Hospital University.

Informed Consent: Informed consent obtained from all human subjects involved in the qualitative component prior to participation.

5. Budget

This study requires a low-to-moderate budget. The primary methodology relies on existing institutional hospital data and open-source computational tools (Python 3, Pyomo, CBC solver), substantially reducing software and data acquisition costs.

Table 3. Estimated Project Budget

Expenditure Category	Type	Estimated Cost
Site visits and transportation (Dammam)	Estimated	500 – 800 SAR
Printing: consent forms, questionnaires, documentation	Estimated	200 – 400 SAR
Research assistance (data entry, transcription), optional	Contingency	500 – 1,500 SAR
Open-access journal publication fee, if applicable	Contingency	1,000 – 3,000 SAR
TOTAL ESTIMATED	—	500 – 5,000 SAR

Funding will be provided through the research support programmes of King Fahad University Hospital and the Undergraduate Research Office at King Fahd University of Petroleum and Minerals (KFUPM), with additional support from the Department of Industrial and Systems Engineering. No external commercial funding is involved.

6. Bibliographical References

- Cardoen B, Demeulemeester E, Beliën J. Operating room planning and scheduling: a literature review. *Eur J Oper Res.* 2010;201(3):921–932.
- Guinet A, Chaabane S. Operating theatre planning. *Int J Prod Econ.* 2003;85(1):69–81.
- Persson M, Persson JA. Health economic modeling to support surgery management at a Swedish hospital. *Omega.* 2009;37(4):853–863.
- May JH, Spangler WE, Strum DP, Vargas LG. The surgical scheduling problem: current research and future opportunities. *Prod Oper Manag.* 2011;20(3):392–405.
- Schouten AM, Flipse SM, van Nieuwenhuizen KE, Jansen FW, van der Eijk AC, van den Dobbelsteen JJ. Operating room performance optimization metrics: a systematic review. *J Med Syst.* 2023;47(1):1.
- Vieira BS, Aiex-Hoff M, Andrade CE, Pessoa AA, Subramanian A. Random-key algorithms for optimizing integrated operating room scheduling. *arXiv:2501.10243.* 2025.
- Al Amin M, Ahmed S, Islam MS, Mitra S, Chowdhury S, Moshood TA. A comprehensive review on operating room scheduling and optimization. *Oper Res.* 2024;25(6).
- Nakka S, Kumar S. Bridging the gap: systems engineering and surgical workflows. *Int J Res Publ Rev.* 2025;16:77–83.
- Pulido R, Aguirre AM, Ortega-Mier M, García-Sánchez Á, Méndez CA. Managing daily surgery schedules in a teaching hospital: a mixed-integer optimization approach. *BMC Health Serv Res.* 2014;14:544.
- Spratt B, Kozan E. An integrated rolling horizon approach to increase operating theatre efficiency. *J Sched.* 2021;24(1):3–25.

11. Celik B, Gul S, Taskin ZC, Akturk MS. A stochastic programming approach to surgery scheduling under parallel processing principle. *Omega*. 2023;115:102799.
12. Addis B, Carello G, Tanfani E. Evaluating the impact of the level of robustness in operating room scheduling problems. *Healthcare (Basel)*. 2024;12(20):2023.
13. Lin YK, Li MY. Solving operating room scheduling problem using artificial bee colony algorithm. *Healthcare (Basel)*. 2021;9(2):152.
14. Vieira BS, Andrade CE, Pessoa AA, Subramanian A. Random-key optimizer framework for integrated operating room scheduling. *Comput Oper Res*. 2025 [In press; arXiv:2501.10243].
15. Vladu A, Ghitea TC, Daina LG, Tîrt DP, Daina MD. Enhancing operating room efficiency: the impact of computational algorithms on surgical scheduling and team dynamics. *Healthcare (Basel)*. 2024;12(19):1906.
16. Al-Araidah O, Momani A, Alawneh F. A resilient framework for operating room management: integrating capacity planning and equitable scheduling. *Ann Oper Res*. 2025. <https://doi.org/10.1007/s10479-025-06565-0>.
17. Amin MA, Ahmed S, Islam MS, Mitra S, Chowdhury S, Moshood TA. Exploring the landscape of operating room scheduling: bibliometric analysis of recent advancements and future prospects. *Biomed Eng Comput Biol*. 2024;16:11795972241271549.
18. Pappada SM, Papadimos TJ, Khuder S, Mack ST, Beachy PZ, Casabianca AB. Contributing factors to operating room delays identified from an electronic health record: a retrospective study. *Anesthesiol Res Pract*. 2022;2022:8635454.
19. Kothe M. Operating room management: scheduling and OR efficiency [Internet]. PerfectServe; 2021 Nov 16 [cited 2026 Apr 18]. Available from: <https://www.perfectserve.com/blog/operating-room-management-scheduling-and-or-efficiency/>
20. Alzahrani MF, Alsubaie AS, Alnasser MS, Aljuaid KA, Alwagdani A, Alharbi HA, et al. Improving operating room efficiency by extending working hours: a hospital-based study. *Cureus*. 2025;17(1):e82582.

7. Appendices

Appendix A: Survey Instruments and Data Collection Materials

Full survey questionnaires, semi-structured interview guides, field observation instruments, and data collection tables are available in the project repository:

[The Interview Survey - Validated](#)

Evaluation of Operating Room Scheduling Practices at King Fahad University Hospital

This survey aims to understand the current state of operating room scheduling from the perspective of surgical practitioners, identify challenges, and gather suggestions for improvement based on clinical experience.

You are being invited to participate in a survey titled:

"Evaluation of Operating Room Scheduling Practices at King Fahad University Hospital."

The purpose of this survey is to gather insights from surgical department staff to identify current challenges and improvement opportunities in the hospital's operating room (OR) scheduling system. Your responses will help inform future optimization projects that aim to increase efficiency, safety, and satisfaction.

Participation is **voluntary**, and your responses will be treated with strict confidentiality. You may choose to skip any question or withdraw at any time. The survey will take approximately **5–7 minutes** to complete.

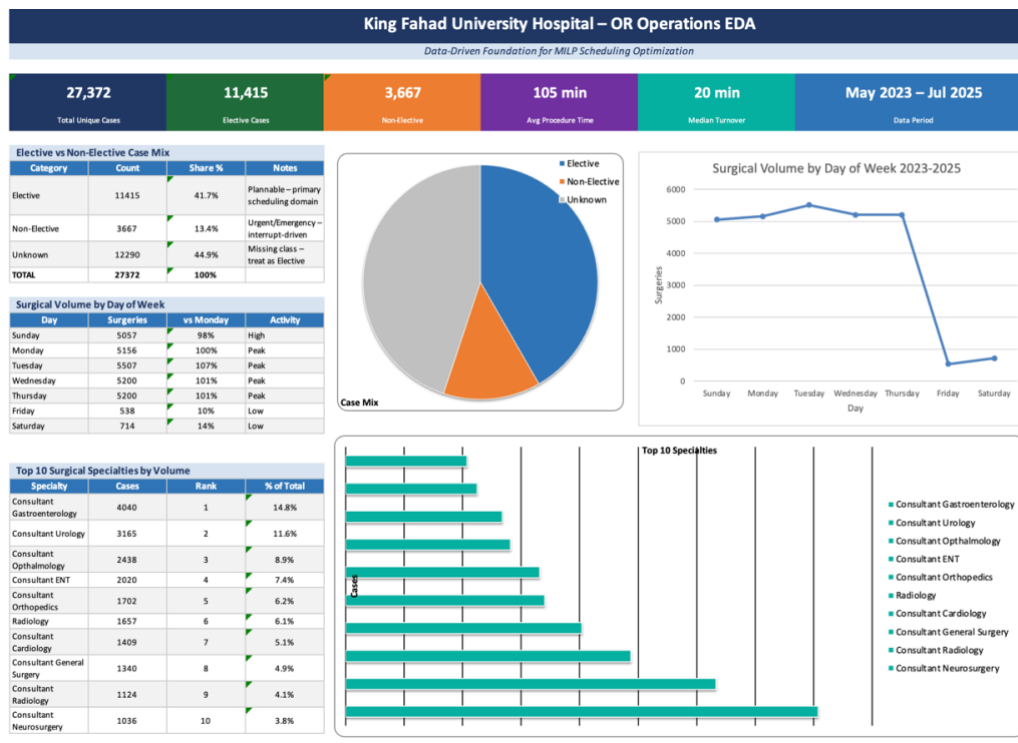
By clicking **"I Agree"** below and continuing the survey, you indicate that:

- You have read and understood the purpose of the survey.
- You consent to participate voluntarily.
- You understand that your responses will be used for research and process improvement purposes only.

If you have any questions or concerns about this study, you may contact the research team via:

Abdulmohsen.alshayeb@gmail.com

0590933943

KFHU OR Book Log - EDAOR MILP Python Code

OR SCHEDULING OPTIMIZATION SYSTEM – KFUH

Based on: Alshayeb et al. (2025) MILP Framework

Pure Python · MILP-inspired greedy optimizer + constraint engine

Appendix B: Multi-Centre Extension and Future Research Directions

This study is designed as a foundational single-centre investigation at KFUH. A subsequent phase is planned as a multi-centre study in collaboration with Imam Abdulrahman Bin Faisal University (IAU) and King Fahd University of Petroleum and Minerals (KFUPM), enabling cross-institutional benchmarking and broader generalisability across the KSA health system. Future research priorities identified by this study include: (i) integration of stochastic surgery duration distributions to improve schedule resilience against intraday variability; (ii) a rolling-horizon extension for dynamic intraday re-scheduling in response to disruptions; (iii) incorporation of surgeon fatigue and workload equity constraints aligned with Liao et al. [15]; and (iv) development of a web-based decision-support interface for direct use by OR scheduling staff at KFUH and partner institutions.

Appendix C: Acknowledgements

The authors extend sincere appreciation to King Saud bin Abdulaziz University for Health Sciences for their support and for providing research training through the 10th Health Professions Conference (HPCX). The authors acknowledge the support of the Undergraduate Research Office at King Fahd University of Petroleum and Minerals (KFUPM) and the Department of Industrial and Systems Engineering for their valuable assistance throughout this research.